

Clément Soubrier

Postdoctoral Fellow in Mathematics

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RESEARCH INTERESTS

Mathematical biology, optimal transport, shape analysis, dynamical systems and stochastic processes applied to cell biology.

ACADEMIC POSITIONS

2026–present **Postdoctoral Fellow**, University of British Columbia, Vancouver.
Supervisor: Khanh Dao Duc. Optimal transport methods for super-resolution biological imaging and high-dimensional genomic data.

EDUCATION

2022–2026 **PhD in Mathematics**, University of British Columbia, Vancouver.
Thesis: *Mathematical modeling and analysis of biological shape heterogeneity and cellular growth*. Supervisor: Khanh Dao Duc.

2018–2022 **MSc, Engineering curriculum, Mathematics**, CentraleSupélec, Université Paris-Saclay.
Maths–Physics and Research track.

2016–2018 **Classe préparatoire (MP)**, Lycée Louis-le-Grand, Paris.

2016 **Baccalauréat and Abitur (AbiBac)**, Lycée Richelieu, Rueil-Malmaison.

PUBLICATIONS AND PREPRINTS

2026 **C. Soubrier**, G. Woollard, A. Warren, K. Dao Duc. *Beyond Procrustes distances: a multilinear Gromov–Wasserstein distance capturing chirality*. Preprint, arXiv:2608.27774.

2025 E. Foxall, **C. Soubrier**. *Time of appearance of a large gap in a dynamic Poisson point process*. Preprint, arXiv:2512.04218.

2025 **C. Soubrier**, A. Madzvamuse, H. A. Eskandarian, K. Dao Duc. *Quantifying the organization and dynamics of *M. smegmatis* morphology from long-term time-lapse atomic force microscopy*. Preprint, bioRxiv.

2024 **C. Soubrier**, E. Foxall, L. Ciandrini, K. Dao Duc. *Optimal control of ribosome population for gene expression under periodic nutrient intake*. Journal of the Royal Society Interface, doi:10.1098/rsif.2023.0652.

TALKS

May 2026 *Integrating shape dynamics to enhance cell motility analysis*.
BIRS workshop A Roadmap towards Developing Mechanobiochemical Models for Single and Collective Cell Migration through Complex Non-Isotropic Environments, Banff.

Mar. 2026 *The multilinear Gromov–Wasserstein distance and its application to chiral shape comparison*.
Kantorovich Initiative Retreat, UBC, Vancouver.

July 2025 *Modeling the meiotic spindle using a spatial birth-death process*.
SMB Annual Meeting, minisymposium Probability and stochastic processes in biology, Edmonton.

Apr. 2024 *Analyzing morphological properties of *M. smegmatis* using temporal data*.
IAM Annual Retreat, UBC, Vancouver.

Sep. 2023 *End to end pipeline for cell shape dynamics analysis from atomic force microscopy time series*.
BIRS workshop Mathematical Methods for Exploring and Analyzing Morphological Shapes across Biological Scales, Banff.

Jan. 2023 *Impact of ribosome population control on the dynamics of protein expression*.
Joint Mathematics Meetings, Boston.

TEACHING

UBC MATH 100, MATH 180, MATH 406, MATH 307, MATH 253

AWARDS

2022–2026 Four year Fellowship, UBC
2021 Globalink Research Internship Award, Mitacs

LANGUAGES

- French (native)
- English (fluent)
- German (fluent)